

Derry Tanti Wijaya

Associate Professor of Data Science · Computational Linguistics & Natural Language Processing

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3,557+ Citations (Google Scholar)	\$1.2M+ Research Funding Raised (PI/Co-PI)	9 PhD Students Supervised	60+ Peer-Reviewed Publications
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Publication venues include ICML, ICLR, NeurIPS, ACL, NAACL, EMNLP, and COLING. h-index available on Google Scholar.

PROFILE

Computer scientist and NLP researcher specializing in **multilingual, multimodal, and multicultural language technologies**. Ph.D. from Carnegie Mellon University (advisor: Tom Mitchell); former Assistant Professor of Computer Science at Boston University and postdoctoral researcher at the University of Pennsylvania (advisor: Chris Callison-Burch). Research spans **mechanistic interpretability and steerability of language models, model editing, evaluation frameworks and LLM-as-a-judge, and data-efficient adaptation for low-resource languages**, with a parallel program building culturally grounded resources and benchmarks (indigenous scripts, honorifics, code-switching) and applying NLP to societal challenges including bias, polarization, toxicity, and misinformation. Author of work at **ICML, ICLR, NeurIPS, ACL, NAACL, and EMNLP**; principal or co-investigator on grants from Google, DARPA, the Open Society Foundation, Danida, and others.

RESEARCH INTERESTS

- **Interpretable & steerable LMs:** mechanistic interpretability of multilingual/multimodal models, safe model editing, controllability, hallucination reduction.
- **Evaluation of generative models:** meta-metrics, LLM-as-a-judge, rubric-agnostic reward reasoning, alignment with human preferences.
- **Low-resource & multilingual NLP:** data-efficient transfer (circuit-targeted fine-tuning), machine translation from dictionaries and grammars, code-switching generation.
- **Multicultural AI & cultural heritage:** culturally aware large multimodal models, preservation of indigenous scripts, honorific systems.
- **NLP for society:** bias, polarization, toxicity, hate speech, and misinformation; computational framing analysis; AI for health.

ACADEMIC APPOINTMENTS

Associate Professor, Data Science — Monash University, Indonesia Program Coordinator, Data Science (2024–present). Co-Director, Monash Data & Democracy Research Hub (2023–present).	2022 – present
Assistant Professor, Computer Science — Boston University Now Adjunct. Co-Director, AI and Education Initiative, Hariri Institute (2021–2022).	2018 – 2022
Postdoctoral Researcher, Computer & Information Science — University of Pennsylvania Advisor: Chris Callison-Burch.	2016 – 2018

EDUCATION

Ph.D., Language Technologies Institute — Carnegie Mellon University Advisor: Tom Mitchell. Thesis: VerbKB — A Knowledge Base of Verbs for Natural Language Understanding.	2016
M.Sc., School of Computing — National University of Singapore Advisor: Stephane Bressan.	2010
B.Comp., School of Computing — National University of Singapore	2004

AWARDS & FELLOWSHIPS

- **Best Data Researcher, Data Science Indonesia Award** (2021)
- **University Provost's Career Development Professorship, Boston University** (2019)
- **Anita Borg Scholarship, Google** (2013)
- **International Science & Technology Fellowship, Fulbright** (2010–2016)

SELECTED PUBLICATIONS

Name in bold. See Google Scholar for the complete list of 60+ works (3,557+ citations).

Interpretability, Model Editing & Evaluation

- Do Language Models Track Entities Across State Changes?
Z. Tang, Q. Zhao, G. Franco, **D. Wijaya**, A. Mueller, S. Schuster, N. Kim
[ICML 2026](#)
- Mechanisms are Transferable: Data-Efficient Low-Resource Adaptation via Circuit-Targeted Supervised Fine-Tuning
K. Nur'aini, A. Purwarianti, A. F. Aji, **D. Wijaya**
[Preprint \(in submission\)](#) [\[link\]](#)
- mR3: Multilingual Rubric-Agnostic Reward Reasoning Models
D. Anugraha, S-Y. Hung, Z. Tang, A. E-S. Lee, **D. Wijaya**, G. I. Winata
[ICLR 2026](#) [\[link\]](#)
- MetaMetrics: Calibrating Metrics for Generation Tasks Using Human Preferences
G. I. Winata, D. Anugraha, L. Susanto, G. Kuwanto, **D. Wijaya**
[ICLR 2025](#) [\[link\]](#)
- Deductive Closure Training of Language Models for Coherence, Accuracy, and Updatability
A. F. Akyürek, E. Akyürek, L. Choshen, **D. Wijaya**, J. Andreas
[ACL Findings 2024](#) [\[link\]](#)
- DUnE: Dataset for Unified Editing
A. F. Akyürek, E. L. Pan, G. Kuwanto, **D. Wijaya**
[EMNLP 2023](#) [\[link\]](#)
- RL4F: Generating Natural Language Feedback with Reinforcement Learning for Repairing Model Outputs
A. F. Akyürek, E. Akyürek, A. Kalyan, P. Clark, **D. Wijaya**, N. Tandon
[ACL 2023](#) [\[link\]](#)

Multilingual, Multimodal & Multicultural NLP

- NusaAksara: A Multimodal and Multilingual Benchmark for Preserving Indonesian Indigenous Scripts
M. F. Adilazuarda, M. Wijanarko, L. Susanto, K. Nur'aini, **D. Wijaya**, A. F. Aji
[ACL 2025](#) [\[link\]](#)
- Do Language Models Understand Honorific Systems in Javanese?
M. R. Farhansyah, I. Darmawan, A. Kusumawardhana, G. I. Winata, A. F. Aji, **D. Wijaya**
[ACL 2025](#) [\[link\]](#)
- WorldCuisines: A Massive-Scale Benchmark for Multilingual and Multicultural Visual Question Answering on Global Cuisines
G. I. Winata et al., **D. Wijaya**
[NAACL 2025](#) [\[link\]](#)
- What Do Indonesians Really Need from Language Technology? A Nationwide Survey
M. D. A. Kautsar, L. Susanto, **D. Wijaya**, F. Koto
[EMNLP 2025](#) [\[link\]](#)
- Linguistics Theory Meets LLM: Code-Switched Text Generation via Equivalence Constrained Large Language Models
G. Kuwanto, C. Agarwal, G. I. Winata, **D. Wijaya**
[Workshop on Computational Developmental Linguistics 2026](#) [\[link\]](#)
- Mitigating Translationese in Low-resource Languages: The Storyboard Approach
G. Kuwanto et al., **D. Wijaya**
[LREC-COLING 2024](#) [\[link\]](#)
- Explain-then-Translate: An Analysis on Improving Program Translation with Self-Generated Explanations

Z. Tang, M. Agarwal, A. G. Shypula, B. Wang, **D. Wijaya**, J. Chen, Y. Kim
EMNLP Findings 2023 [\[link\]](#)

NLP for Society: Bias, Polarization, Misinformation & Health

- A Multi-Labeled Dataset for Indonesian Discourse: Examining Toxicity, Polarization, and Demographics
L. Susanto, M. Wijanarko, P. Pratama, Z. Tang, F. Akyas, T. Hong, I. Idris, A. F. Aji, **D. Wijaya**
ACL Findings 2025 [\[link\]](#)
- COVID-19 Vaccine Misinformation in Middle Income Countries
J. Kim, B. R. Bak, A. Agrawal, J. Wu, V. J. Wirtz, T. Hong, **D. Wijaya**
EMNLP 2023 [\[link\]](#)
- Effects of #coronavirus Content Moderation on Misinformation and Anti-Asian Hate on Instagram
T. Hong, Z. Tang, M. Lu, Y. Wang, J. Wu, **D. Wijaya**
New Media & Society 2023 [\[link\]](#)
- Agenda Setting, Cross-Cutting Effects, and Political Expression on Social Media: The Gun Violence Case
L. Guo, Y. Zhang, K. Mays, A. F. Akyürek, **D. Wijaya**, M. Betke
Communication Research 2023 [\[link\]](#)
- Proposing an Open-Sourced Tool for Computational Framing Analysis of Multilingual Data
L. Guo, C. Su, S. Paik, V. Bhatia, V. P. Akavoor, G. Gao, M. Betke, **D. Wijaya**
Digital Journalism 2023 [\[link\]](#)

COMPETITIVE RESEARCH GRANTS

Over \$1.2M USD raised as Principal Investigator or Co-Investigator. Selected:

- **LEARN: Label-Efficient Active Resilient Network** — DARPA (Learning with Less Labels), \$462k (Co-PI)
- **Digitalising Waste Management for a Just Green Transition in Indonesian Smart Cities** — Danida (Denmark), \$300k
- **Contextualizing LLMs for Women's Sexual & Reproductive Health in Indonesia** — Google Research Award, \$60k (PI)
- **Exploring the Evolution of Racial Biases through Framing Analysis** — Google Research Scholar Award, \$60k (PI)
- **IndigiSEA: Indigenous Digital Futures for Southeast Asia** — Monash Incubator Grant, \$53k
- **Boosting Monash's Global Reputation: Data & Democracy Hub** — Monash Incubator Grant, \$53k
- **Monitoring Hate Speech toward Vulnerable Communities, 2024 Indonesia Election** — Open Society Foundation, \$37k
- **Sea-Tech: Sustainable Seaweed Cultivation through Smart Technology** — PAIR (Australia-Indonesia), \$35k
- **Exploring Generative AI for Moderating Online Polarisation** — Monash Data Futures Institute, \$35k (PI)
- **ASEAN Disinformation Index** — Monash Data Futures Institute, \$33k
- **Bhinneka: Multilingual Indonesian Resources** — Indonesian Ministry of Education (US Collaboration), \$30k

RESEARCH SUPERVISION

Doctoral Students (9 total: 4 ongoing, 4 completed, all with top-venue output)

- **Ongoing:** Khumaisa Nur'aini (Monash), Adryan Kusumawardhana (Monash), Zilu Tang (BU), Ge Gao (BU).
- **Tahsina Hashem** (Monash, 2026) — Faithful and Fair Text Generation from Structured and Visual Data.
- **Muhammed Yusuf Kocyigit** (BU, 2025) — Improving Evaluation Practices in NLP: Contamination and Metric Limitations.
- **Afra Feyza Akyürek** (BU, 2025) — Advancing Neural Networks Beyond Initial Training.
- **Isidora Tourni** (BU, 2023) — Neural Machine Translation for Low-Resource Conditions.

Master's Students (selected; many with first-author papers at ACL/ICLR)

- **Garry Kuwanto** (BU, 2026) — MetaMetrics: a Bayesian optimization framework for calibrating LLM evaluation metrics (ICLR 2025).
- **Iwan Darmawan** (Monash, 2024) — Javanese Honorific System corpus (ACL 2025).
- **Fariz Akyas** (Monash, 2024) — Annotator Identity in Indonesian Political Polarization Corpus (ACL Findings 2025).

Full supervision record (15+ Master's students) available on request.

TEACHING

Graduate instructor, Master of Data Science, Monash University Indonesia. Selected courses with student evaluation (SETU) scores:

- **Machine Learning (ITI5201)** — 4.29/5 (2025). Grounds ML in real-world societal challenges.
- **Data Analysis for Semi-Structured Data (ITI5212)** — 4.50/5 (2025), 4.69/5 (2024). Focus on generative AI and LLMs.
- **Master Thesis supervision (ITI5126-5229)** — 4.42/5. Students published and presented at top conferences.
- **Data Wrangling (ITI5196)** — 4.66/5 (2023). Introduction to Data Science (ITI5145) — 4.08/5.

LEADERSHIP & PROFESSIONAL SERVICE

Academic Leadership

- **Program Coordinator, Data Science**, Monash University Indonesia (2024–present).
- **Co-Director, Monash Data & Democracy Research Hub** (2023–present) — partnerships across government, academia, and civil society; convened the Data & Democracy Dialogue (Jakarta 2024; Kuala Lumpur 2025).
- **Faculty of IT Course Governance, Quality & Resourcing Sub-committee** (2026–present); Education Committee (2024–present); Research Committee (2024–2025).
- **Co-Director, AI & Education Initiative**, Hariri Institute, Boston University (2021–2022).

Community Service to the Field

- **Area Chair / Senior Area Chair / Session Chair** for ACL conferences, ICLR, and NeurIPS (ongoing).
- **Organizer**: AACL 2026 (Hengqin); 2nd Workshop on Southeast Asian Language Processing (COLING 2025); 7th Workshop on Computational Approaches to Linguistic Code-Switching (NAACL 2025); AACL 2023 (Bali).

References available upon request.